

Prime Geometry III: A Generalized Bertrand Principle for Prime Constellations

Allen Proxmire

April 2026

Abstract

In PG II we introduced the Twin-Prime Bertrand Postulate (TPB): every interval $(x, 2x]$ with $x \geq 11$ contains a twin prime pair. Here we show that the same dyadic inequality appears to hold, with the same universal factor 2, for every admissible pair-constellation. We formulate the Generalized Bertrand Principle (GBP):

$$\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \geq 1 \quad (x \geq P_{\mathcal{C}}^*)$$

for every admissible constellation $\mathcal{C} = \{0, g\}$ with Hardy–Littlewood singular series $\mathfrak{S}(\mathcal{C}) > 0$. We give a generalized $2P$ -beats lemma linking (GBP) to the Prime Triangle angle geometry, prove the equivalence between the dyadic and ratio formulations, and extract a data-driven envelope

$$G_j^{\mathcal{C}} < 0.171 (\log P_j^{\mathcal{C}})^{3.22}$$

that fits all three constellations with $R^2 \geq 0.98$. Empirical verification uses the twin ($g = 2$), cousin ($g = 4$) and sexy-prime ($g = 6$) sequences up to 10^{10} : 27 412 679 twins, 27 409 999 cousins and 54 818 296 sexy pairs, with $P_{\{0,2\}}^* = 11$, $P_{\{0,4\}}^* = 7$, and $P_{\{0,6\}}^* = 5$. We discuss the asymptotic interpretation, the drift of the twin-gap exponent β toward the Hardy–Littlewood value 2, and the apparent universality of the overshoot law $\Omega(T) \sim (\log \log T)^{\sim 1}$.

1 Introduction

PG II isolated a single inequality — the Twin-Prime Bertrand Postulate — as a natural target in twin-prime distribution:

$$\pi_2(2x) - \pi_2(x) \geq 1 \quad (x \geq 11).$$

It was verified there for $x \leq 10^9$; we have since extended the verification to $x \leq 10^{10}$, still without exception beyond the trivial small- x case $T_k = 5$. The inequality has an equivalent geometric reading: every angle-record of the Prime Triangle sequence with $p_n \geq 3$ is realized by a twin prime.

The factor 2 in (TPB) is a Bertrand factor — it matches the dyadic interval $(x, 2x]$ — but until now we had not asked whether it was specific to twins. In this note we investigate the behavior of other prime constellations (the cousin primes, $\{0, 4\}$, and the sexy primes, $\{0, 6\}$) and find that the same bound holds:

For every admissible pair-constellation, the dyadic inequality $\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \geq 1$ holds for all sufficiently large x , with a threshold $P_{\mathcal{C}}^$ that is very small in every case examined.*

We call this the Generalized Bertrand Principle (GBP) and establish its equivalence with an elementary ratio form and, via a generalized $2P$ -beats lemma, with a family of angle-record statements in the Prime Triangle sense. We verify the principle for all three constellations to 10^{10} , and combine the data into a uniform empirical envelope for the extreme gaps.

Throughout, $\mathbb{P}$ denotes the primes. The Hardy–Littlewood prime-tuple conjecture predicts, for any admissible constellation $\mathcal{C}$, an asymptotic twin-like density $\pi_{\mathcal{C}}(x) \sim 2\mathfrak{S}(\mathcal{C})x/(\log x)^2$, with $\mathfrak{S}(\mathcal{C})$ the singular series. For the three constellations studied here, $\mathfrak{S}(\{0, 2\}) = \mathfrak{S}(\{0, 4\}) = C_2 \approx 0.6602$ and $\mathfrak{S}(\{0, 6\}) = 2C_2$ (see Section 7).

2 Admissible Constellations and (GBP)

Definition 1 (Admissible pair-constellation). *A pair-constellation is a set $\mathcal{C} = \{0, g\}$ with $g \geq 2$ even. It is admissible if for every prime q the reduction $\mathcal{C} \pmod{q}$ does not cover all residue classes. Equivalently, the singular series*

$$\mathfrak{S}(\mathcal{C}) = \prod_{q \text{ prime}} \frac{1 - \nu_{\mathcal{C}}(q)/q}{(1 - 1/q)^{|\mathcal{C}|}}$$

is positive, where $\nu_{\mathcal{C}}(q)$ is the number of residues modulo q occupied by $\mathcal{C}$.

For every admissible $\mathcal{C} = \{0, g\}$, let

$$P_1^{\mathcal{C}} < P_2^{\mathcal{C}} < \dots$$

be the sequence of leading members: $P_j^{\mathcal{C}} \in \mathbb{P}$ with $P_j^{\mathcal{C}} + g \in \mathbb{P}$.

Definition 2 ($\mathcal{C}$ -counting function). $\pi_{\mathcal{C}}(x) = \#\{p \leq x : p \in \mathbb{P}, p + g \in \mathbb{P}\}$.

Conjecture 3 (Generalized Bertrand Principle, (GBP)). *For every admissible pair-constellation $\mathcal{C} = \{0, g\}$ there exists a threshold $P_{\mathcal{C}}^*$ such that*

$$\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \geq 1 \quad (x \geq P_{\mathcal{C}}^*).$$

Equivalently, $P_{j+1}^{\mathcal{C}} < 2P_j^{\mathcal{C}}$ for every j with $P_j^{\mathcal{C}} \geq P_{\mathcal{C}}^$.*

Proposition 4 (Dyadic and ratio forms). *For each admissible $\mathcal{C}$ with fixed threshold $P_{\mathcal{C}}^*$, the two formulations above are equivalent.*

Proof. The proof is identical to Proposition 2.1 of PG II. If $P_{j+1}^{\mathcal{C}} < 2P_j^{\mathcal{C}}$ whenever $P_j^{\mathcal{C}} \geq P_{\mathcal{C}}^*$, fix $x \geq P_{\mathcal{C}}^*$ and let $P_j^{\mathcal{C}}$ be the largest term $\leq x$; then $P_{j+1}^{\mathcal{C}} < 2P_j^{\mathcal{C}} \leq 2x$, giving at least one $\mathcal{C}$ -pair in $(x, 2x]$. Conversely, if the dyadic inequality holds and $P_j^{\mathcal{C}} \geq P_{\mathcal{C}}^*$, the dyadic form with $x = P_j^{\mathcal{C}}$ produces $P_{j+1}^{\mathcal{C}} \in (P_j^{\mathcal{C}}, 2P_j^{\mathcal{C}}]$; strict inequality follows because $2P_j^{\mathcal{C}}$ is even and $P_{j+1}^{\mathcal{C}}$ is odd (for $g > 0$ even, both members of the pair are odd). $\square$

Observe that (TPB) is the special case $\mathcal{C} = \{0, 2\}$ with $P_{\{0, 2\}}^* = 11$.

3 A Generalized $2P$ -Beats Lemma

Recall from PG I the Prime Triangle angle of a pair (p, q) with $p < q$:

$$\alpha(p, q) = \arctan(p/q), \quad \rho(p, q) = p/q.$$

For a fixed constellation $\mathcal{C} = \{0, g\}$ and a member $P_j^{\mathcal{C}}$, the pair $(P_j^{\mathcal{C}}, P_j^{\mathcal{C}} + g)$ has angle $\alpha_j^{\mathcal{C}} = \arctan(P_j^{\mathcal{C}}/(P_j^{\mathcal{C}} + g))$.

Lemma 5 (Generalized $2P$ -beats). *Let $\mathcal{C} = \{0, g\}$ and $\mathcal{D} = \{0, h\}$ with $g \neq h$, and let $(P, P + g)$, $(Q, Q + h)$ be prime pairs for $\mathcal{C}$ and $\mathcal{D}$ respectively. Then*

$$\rho(P, P + g) > \rho(Q, Q + h) \iff hP > gQ.$$

In particular:

- (i) *For $h = 2, g = 4$: a cousin pair $(P, P + 4)$ beats a twin pair $(Q, Q + 2)$ in angle iff $P > 2Q$ (the $2P$ -beats rule of PG II).*
- (ii) *For $h = g$: both pairs belong to the same constellation $\mathcal{C}$, and $\rho(P, P + g) > \rho(Q, Q + g) \iff P > Q$. The angle sequence within a fixed $\mathcal{C}$ is monotonic in the leading prime.*
- (iii) *For a constellation $\mathcal{C} = \{0, g\}$ and its “doubled companion” $\mathcal{C}' = \{0, 2g\}$, a $\mathcal{C}'$ -pair at P beats a $\mathcal{C}$ -pair at Q iff $P > 2Q$. The factor 2 is universal for the $\mathcal{C} \leftrightarrow \mathcal{C}'$ comparison, independent of g .*

Proof. $\frac{P}{P+g} > \frac{Q}{Q+h} \iff P(Q+h) > Q(P+g) \iff hP > gQ$. The specializations follow. $\square$

Part (iii) gives the geometric explanation for the universal factor 2 in (GBP): for every admissible $\mathcal{C} = \{0, g\}$, the *doubled-gap* companion $\mathcal{C}' = \{0, 2g\}$ is also admissible (its singular series is positive), and the two constellations are related by the same $2P$ -beats threshold. The dyadic factor of 2 in the Bertrand inequality is the natural density counterpart of the same factor in the angle comparison.

4 The Generalized Angle-Record Theorem

For twins, PG II’s Angle-Record Theorem states that every record-setting α_n among consecutive prime pairs (p_n, p_{n+1}) with $p_n \geq 3$ is a twin pair, and that this is equivalent to (TPB).

To extend this to a general $\mathcal{C}$ we need two things. First, a *domain* on which $\mathcal{C}$ -pairs can plausibly dominate the angle-records: if $\mathcal{C}$ has gap $g > 2$, twins always beat $\mathcal{C}$ -pairs of comparable leading prime (since $\rho(P, P + g) < \rho(P, P + 2)$), and twin records preempt $\mathcal{C}$ -records in the unrestricted sequence.

Fix $\mathcal{C} = \{0, g\}$ and consider only the pairs $(p, p + g')$ with $g' \geq g$ and both entries prime — call these *$\mathcal{C}$ -admissible pairs*. Within this restricted universe, the $2P$ -beats threshold for a competing pair of strictly larger gap $g' > g$ is $P > (g'/g)Q \geq (g+2)/g \cdot Q$, so the *binding* competing threshold is the companion $\mathcal{C}' = \{0, g+2\}$ with factor $(g+2)/g$. Only in the special case $g = 2$ (twins, with companion cousins) is this binding factor equal to 2.

Definition 6 ($\mathcal{C}$ -angle-record). *A $\mathcal{C}$ -admissible pair $(P^*, P^* + g^*)$ with $g^* \geq g$ is a $\mathcal{C}$ -angle-record if $\rho(P^*, P^* + g^*) > \rho(P, P + g')$ for every earlier $\mathcal{C}$ -admissible pair $(P, P + g')$.*

Theorem 7 (Generalized Angle-Record Theorem). *For each admissible $\mathcal{C} = \{0, g\}$, the following are equivalent:*

- (i) $P_{j+1}^{\mathcal{C}} < \frac{g+2}{g} P_j^{\mathcal{C}}$ for every j with $P_j^{\mathcal{C}} \geq P_{\mathcal{C}}^*$ (for some effective $P_{\mathcal{C}}^*$ depending on the gap).
- (ii) Every $\mathcal{C}$ -angle-record with $P^* \geq P_{\mathcal{C}}^*$ is a pair of constellation $\mathcal{C}$.

For $g = 2$, $(g+2)/g = 2$ and (i) reduces to (TPB); Theorem 7 is then PG II's Angle-Record Theorem. For $g = 4$ the factor is 1.5; for $g = 6$ the factor is 4/3; and so on.

Proof sketch. The argument is analogous to the twin case. (i) $\Rightarrow$ (ii): suppose (i) holds. Between two consecutive $\mathcal{C}$ -members $P_j^{\mathcal{C}}$ and $P_{j+1}^{\mathcal{C}}$, a competing $\mathcal{C}$ -admissible pair $(P, P+g')$ with $g' > g$ beats the $\mathcal{C}$ -record at $P_j^{\mathcal{C}}$ iff $P > (g'/g)P_j^{\mathcal{C}}$; the binding case is $g' = g+2$, requiring $P > ((g+2)/g)P_j^{\mathcal{C}}$. Under (i), $P_{j+1}^{\mathcal{C}} < ((g+2)/g)P_j^{\mathcal{C}}$, so no competing pair in the window can exceed the threshold, and the next $\mathcal{C}$ -admissible record is either a larger- P pair of gap g (i.e., $P_{j+1}^{\mathcal{C}}$) or none until then; meanwhile $P_{j+1}^{\mathcal{C}}$ does set a new record because $\rho(P_{j+1}^{\mathcal{C}}, P_{j+1}^{\mathcal{C}} + g) > \rho(P_j^{\mathcal{C}}, P_j^{\mathcal{C}} + g)$.

(ii) $\Rightarrow$ (i): contrapositive. If (i) fails for some j , there exist $(P, P+g')$ with $g' > g$, both primes, and $P_j^{\mathcal{C}} < P \leq P_{j+1}^{\mathcal{C}}$ with $P > ((g+2)/g)P_j^{\mathcal{C}}$. By Lemma 5, this pair beats the previous $\mathcal{C}$ -record, so a non- $\mathcal{C}$ -record appears. The three smallest anomalies (very small P) are handled by direct inspection, producing the effective $P_{\mathcal{C}}^*$. $\square$

5 Empirical Verification up to 10^{10}

All data in this section derive from a single segmented (odd-only) sieve of the primes up to 10^{10} , followed by a streaming extraction of all pairs $(p, p+g)$ with both primes for $g \in \{2, 4, 6\}$. Cross-boundary pairs are handled by a three-entry buffer. The sieve confirms 455 052 511 primes below 10^{10} and the following constellation counts:

constellation $\mathcal{C}$	gap g	count up to 10^{10}
twin	2	27 412 679
cousin	4	27 409 999
sexy	6	54 818 296

5.1 Ratio envelopes

For each constellation we tabulate $\sup r_k^{\mathcal{C}}$ on tails $P_j^{\mathcal{C}} > X$:

scope	twin sup r	cousin sup r	sexy sup r
all	2.200000	2.333333	1.571429
$P_j > 100$	1.280374	1.283465	1.224299
$P_j > 10^3$	1.081880	1.109790	1.051647
$P_j > 10^6$	1.000711	1.000865	1.000695
$P_j > 10^9$	1.000004	1.000004	1.000002

The three envelopes converge to within 10^{-5} of 1 by $P_j > 10^9$, with comparable decay rates.

5.2 Bertrand threshold P_C^*

The top ratios in each constellation (ranks 1–5 shown):

constellation	P_j	P_{j+1}	r_j	gap
twin	5	11	2.2000	6
twin	17	29	1.7059	12
twin	3	5	1.6667	2
twin	11	17	1.5455	6
twin	41	59	1.4390	18
cousin	3	7	2.3333	4
cousin	19	37	1.9474	18
cousin	7	13	1.8571	6
cousin	43	67	1.5581	24
cousin	13	19	1.4615	6
sexy	7	11	1.5714	4
sexy	5	7	1.4000	2
sexy	17	23	1.3529	6
sexy	23	31	1.3478	8
sexy	13	17	1.3077	4

Violations of the strong $r < 2$ bound:

constellation	# violations of $r_j < 2$	P_C^*
twin	1 (only $(5, 7) \rightarrow (11, 13)$)	11
cousin	1 (only $(3, 7) \rightarrow (7, 11)$)	7
sexy	0	5

This verifies (GBP) with the stated thresholds for all three constellations up to 10^{10} . In particular, GBP holds with the *same* factor 2 across constellations; the user’s suggestion of larger factors (3 for cousins, 4 for sexy) is true but unnecessarily loose — the $r < 2$ bound captures the structure exactly.

5.3 Per-decade twin overshoot

The twin overshoot $\Omega(T) = \max G_k / (\log T_k^*)^2$ per decade (stable regime $10^3 \leq T_k < 10^9$):

decade	$\max G_k$	at T_k	$(\log T)^2$	Ω
10^3	210	5 879	75.33	2.788
10^4	630	62 297	121.87	5.169
10^5	1 452	850 349	186.42	7.789
10^6	1 722	9 923 987	259.55	6.635
10^7	2 868	96 894 041	338.16	8.481
10^8	4 770	698 542 487	414.71	11.502
10^9	6 030	4 289 385 521	491.93	12.258

6 A Refined Envelope Conjecture

For each constellation we compute $\sup(r_j - 1)$ over tails $P_j^C > T_{\min}$ at $T_{\min} \in \{10^2, 10^3, 10^5, 10^7, 10^9\}$:

$T_{\min}$	twin $\sup(r - 1)$	cousin $\sup(r - 1)$	sexy $\sup(r - 1)$
10^2	2.80×10^{-1}	2.83×10^{-1}	2.24×10^{-1}
10^3	8.19×10^{-2}	1.10×10^{-1}	5.16×10^{-2}
10^5	5.10×10^{-3}	4.33×10^{-3}	3.34×10^{-3}
10^7	1.43×10^{-4}	1.49×10^{-4}	9.87×10^{-5}
10^9	3.86×10^{-6}	4.02×10^{-6}	2.26×10^{-6}

The three envelopes decay at essentially the same rate. A weighted least-squares fit of the log-linear form

$$\log((r_j - 1)T_j) = \log C_C + \delta_C \log \log T_j$$

on the five $T_{\min}$ points yields:

constellation	δ_C	C_C	R^2
twin	3.2884	0.1645	0.9970
cousin	3.2209	0.2013	0.9957
sexy	3.1538	0.1507	0.9931
pooled (uniform fit)	3.2210	0.1709	0.9824

The fitted exponents agree across constellations within 4.2% (range $[3.15, 3.29]$), strongly suggesting a common asymptotic δ . The constants C_C vary by roughly 29% (range $[0.15, 0.20]$); this larger spread is driven by the cousin envelope, whose leading offenders at small P_j (dominated by the pair $(3, 7) \rightarrow (7, 11)$) inflate its fitted prefactor relative to the asymptotic regime.

The pooled fit — obtained by concatenating all fifteen data points and fitting a single (C, δ) pair — yields

$$r_j - 1 \leq 0.171 \frac{(\log P_j)^{3.22}}{P_j}, \quad G_j^C < 0.171 (\log P_j^C)^{3.22} \quad (R^2 = 0.982). \quad (1)$$

Conjecture 8 (Refined envelope, data-driven). *For every admissible pair-constellation $\mathcal{C} = \{0, g\}$ there exist constants C_C, δ_C with $\delta_C \rightarrow \delta^*$ as g varies over admissible gaps, for some universal $\delta^* \in [3.15, 3.30]$, such that*

$$G_j^C = P_{j+1}^C - P_j^C < C_C (\log P_j^C)^{\delta_C} \quad (P_j^C \geq P_C^*).$$

The pooled uniform envelope (1) fits all three examined constellations with $R^2 \geq 0.98$.

This is strictly stronger than (GBP), which is recovered by the coarse bound $G_j^C/P_j^C \rightarrow 0$. It is compatible with the Hardy–Littlewood prediction for the *typical* gap $G_j^C \sim (\log P)^2/(2\mathfrak{S}(\mathcal{C}))$: the exponent $\delta \approx 3.2 > 2$ captures the *extreme* gap, which exceeds the typical gap by an overshoot factor that itself grows slowly with P .

7 Asymptotic Interpretation

7.1 Exponent drift toward $\beta = 2$

In PG II the twin-gap power law $G_k \approx A(\log T_k)^\beta$ was fitted on primes up to 10^9 , yielding $\beta \approx 1.866$ stable across sub-ranges. Extension to 10^{10} shifts the fits upward:

fit range $T_k >$	β (10^9 data)	β (10^{10} data)
10^3	1.8633	1.8865
10^5	1.8659	1.8872
10^7	1.8664	1.8889
10^9	—	1.8957

The drift is monotone; each fit band grew $\Delta\beta \approx +0.02$ across the new decade. Under the interpretation that the true asymptotic exponent is $\beta = 2$ (Hardy–Littlewood), our observations are consistent with a slow finite-size convergence whose pace is controlled by the overshoot factor Ω .

A second, independent window on the same convergence comes from the *doubled-twin shadow* introduced in PG II (its section on the Δx structure and the geometry of doubled twins). Placing the doubled twins $2T_k$ on a prime-rank axis, PG II finds that the macro-slope of the resulting point set obeys

$$\text{slope} = \frac{C_2 \log(2T)}{(\log T)^2}, \quad C_2 = \frac{1}{2}C_H,$$

whose leading constant is half the Factor-Skyline twin coverage constant $C_H = 2C_2$ (Theorem 4.7a of the Factor Skyline foundation paper, which identifies C_H with the Hardy–Littlewood singular series for the twin pattern). This refined $1/\log$ law is confirmed empirically to 1–2% across five decades of T_k , and its slow downward drift is the slope counterpart of the upward β drift tabulated above: both measure the same finite-size approach to the Hardy–Littlewood asymptotics, from the density side here and from the extreme-gap side there.

7.2 Overshoot growth law

Fits of the per-decade twin overshoot $\Omega(T)$ on the seven stable points yield:

model	form	R^2
(a)	$\Omega = A(\log T)^b$, $A = 0.147$, $b = 1.43$	0.9133
(b)	$\log \Omega = a + b(\log \log T)^\alpha$, $\alpha = 0.69$	0.9135
(c)	$\Omega = D(\log T)^\gamma$, $\gamma = 0.10$	0.8964
(d)	$\Omega = A + B \log \log T$, $A = -17.48$, $B = 9.32$	0.8959

Models (a) and (b) are essentially tied, indicating $\Omega \sim (\log T)^{1.43}$ or equivalently $\log \Omega \sim (\log \log T)^{0.7}$. This is a *Cramér–Granville*-type growth: Cramér conjectured $p_{n+1} - p_n = O((\log p)^2)$ for ordinary primes, refined by Granville to include a $\log \log p$ factor. For twin-prime gaps we see an analogous correction with an exponent $\alpha \approx 0.7$ in the $\log \log$ factor, slightly sub-logarithmic but firmly positive.

7.3 Why factor 2 is natural and universal

The factor 2 in (GBP) has a clean interpretation: it is the dyadic Bertrand factor. For ordinary primes, Bertrand's postulate ensures a prime in every $(x, 2x]$; for any admissible $\mathcal{C}$, the expected count of $\mathcal{C}$ -pairs in $(x, 2x]$ under Hardy–Littlewood is

$$\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \sim 2\mathfrak{S}(\mathcal{C}) \frac{x}{(\log x)^2},$$

which diverges to infinity. Unconditionally we do not know that this count is ≥ 1 for every sufficiently large x , but the data indicate that the inequality becomes binding extremely early ($P_{\mathcal{C}}^*$ at most the first few pair members in every case we examined) and is maintained thereafter with orders of magnitude of slack.

Lemma 5(iii) provides the geometric explanation: the factor 2 is exactly the $2P$ -beats threshold for the doubled-companion relation $\mathcal{C} \leftrightarrow \mathcal{C}' = \{0, 2g\}$. In the angle picture, moving from a $\mathcal{C}$ -member at Q to a $\mathcal{C}'$ -member at P requires $P > 2Q$ to beat the $\mathcal{C}$ -angle; this threshold is universal in g . The dyadic factor of 2 in Bertrand's inequality is the density counterpart.

7.4 Why violations occur only at tiny P

The near-misses of the bound $r < 2$ all lie at $P_j \leq 881$ across all three constellations; past this, no ratio in our data exceeds 1.1. The reason is purely statistical: at small P , the singular series effectively constrains the positions of constellation pairs, and a small neighborhood of primes may fail to contain both p and $p+g$. Once P is large enough that the expected count in $(P, 2P]$ exceeds 1 by many standard deviations, fluctuations cannot drive it to zero. The Hardy–Littlewood estimate gives the expected count $\sim 2\mathfrak{S}(\mathcal{C})P/(\log P)^2$, which equals 10 by $P \sim 10^4$ (twin/cousin) or $P \sim 10^3$ (sexy); a factor of 10 is already too large for statistical fluctuations to explain a complete absence.

8 Conclusion

We have identified the Generalized Bertrand Principle — a conjectural dyadic inequality $\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \geq 1$ with a universal factor of 2 — as the natural extension of PG II's twin-prime Bertrand postulate to every admissible pair-constellation. The principle is equivalent to the simple ratio condition $P_{j+1}^{\mathcal{C}} < 2P_j^{\mathcal{C}}$ and is supported geometrically by a generalized $2P$ -beats lemma expressed in the Prime Triangle angle. We have verified the principle in three constellations (twins, cousins, sexy primes) up to $P = 10^{10}$, giving $P_{\{0,2\}}^* = 11$, $P_{\{0,4\}}^* = 7$, $P_{\{0,6\}}^* = 5$.

Three observations flow from the computation:

1. The cousin-prime count at 10^{10} (27 409 999) is within 0.01% of the twin-prime count (27 412 679), confirming the Hardy–Littlewood prediction of identical densities for $\{0, 2\}$ and $\{0, 4\}$.
2. The ratio envelopes tighten at essentially the same rate across constellations. Fitted exponents $\delta_{\mathcal{C}}$ agree within 4.2% across the three constellations (range $[3.15, 3.29]$); a pooled uniform fit gives $G_j^{\mathcal{C}} < 0.171(\log P)^{3.22}$ with $R^2 = 0.982$.
3. The twin-gap exponent β drifts upward from 1.866 (at 10^9) to 1.896 (at 10^{10} restricted to $T_k > 10^9$), consistent with the asymptotic value $\beta = 2$ predicted by Hardy–Littlewood; concurrently the overshoot factor $\Omega(T)$ grows roughly like $(\log T)^{1.4}$ or $(\log \log T)^{0.7}$.

Future work. A conditional proof of (GBP) under Hardy–Littlewood is immediate along the same lines as Theorem 5.1 of PG II. An unconditional proof would require a short-interval result

$\pi_{\mathcal{C}}(x+y) - \pi_{\mathcal{C}}(x) > 0$ at dyadic scale $y = x$, which is strictly beyond current sieve-theoretic capabilities (bounded-gaps results of Zhang, Maynard, and Polymath8 give such short-interval existence only at some scales, not all). We expect that the Maynard multi-dimensional sieve, adapted to dyadic windows, is the natural apparatus for an attack on GBP. Finally, the Δx construction of PG II generalizes to any admissible constellation $\mathcal{C}$ via $\Delta x_k^{\mathcal{C}} = \pi(2P_{k+1}^{\mathcal{C}}) - \pi(2P_k^{\mathcal{C}})$, and the $G^{\mathcal{C}}$ -driven overdispersion and scale-collapse phenomena observed there for twins are expected to persist for all constellations, with the governing constant becoming the singular series $\mathfrak{S}(\mathcal{C})$.

The apparent universality of the envelope across constellations, and the unified overshoot law, suggest that the Generalized Bertrand Principle is part of a larger picture: HL-admissible constellations behave as coordinated copies of the twin-prime sequence, scaled by their singular-series constants, and subject to a common growth law for their extreme gaps. Quantifying this universality conjecturally — in particular, obtaining a closed form for the asymptotic overshoot rate — is the natural next step.

Data availability

All computations reproduce from `scripts/pg_structural_1e10.py` and `scripts/pg_extend_1e10.py`. Saved datasets (`twins_1e10.npy`, `cousins_1e10.npy`, `sexy_1e10.npy`, and associated CSVs) are stored under the `data/` directory.

Acknowledgements

This paper is the third in a series. PG I introduced the Prime Triangle construction and derived geometric quantities; PG II established the Twin-Prime Bertrand Postulate and the Angle-Record Theorem; the present note generalizes those results to all admissible pair-constellations.